

Continuum Mechanics HW2

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Technion

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1 Vorticity

1. Write the definition of ω_{ij} and multiply both sides by ε_{ijl} . Simplify to obtain a single summand.
2. Multiply the original definition by ε_{ijl} and use Levi-Civita tensor identities to obtain

$$\omega = \nabla \times \mathbf{v} \quad (1)$$

1.1 part 1

The definition of the vorticity is

$$\omega_{ij} = -\frac{1}{2}\varepsilon_{ijk}\omega_k \quad (2)$$

multiplying by ε_{ijl}

$$\varepsilon_{ijl}\omega_{ij} = -\frac{1}{2}\varepsilon_{ijl}\varepsilon_{ijk}\omega_k \quad (3)$$

$$\varepsilon_{ijl}\omega_{ij} = -\frac{1}{2}\left(2\delta_{kl} \cdot (-1)^2\right)\omega_k \quad (4)$$

$$\varepsilon_{ijl}\omega_{ij} = -\omega_l \quad (5)$$

1.2 part 2

By original definition I imagine this is what's meant:

$$\omega_{ij} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i}\right) \quad (6)$$

multiplying both sides by ε_{ijl}

$$\varepsilon_{ijl}\omega_{ij} = \frac{1}{2}\varepsilon_{ijl}\left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i}\right) \quad (7)$$

$$= \frac{1}{2}\left(\varepsilon_{ijl}\frac{\partial v_i}{\partial x_j} - \varepsilon_{ijl}\frac{\partial v_j}{\partial x_i}\right) \quad (8)$$

$$= -\varepsilon_{ijl}\frac{\partial v_i}{\partial x_j} \quad (9)$$

combining with the result of part 1 we get

$$\omega_l = \varepsilon_{ijl}\frac{\partial v_i}{\partial x_j} \quad (10)$$

which by definition is

$$\omega = \nabla \times \mathbf{v} \quad (11)$$

2 Streamlines and pathlines

Consider the following 2D flow

$$\mathbf{v}(x, y) = \begin{pmatrix} \alpha y t \\ \beta \end{pmatrix} \quad (12)$$

1. Find the streamlines $\mathbf{x}(x, t)$ and the pathlines $\mathbf{x}(t, \mathbf{x}_0)$ of the flow.
2. Create an animation of the motion of a particle starting at position $(0, 0)$ at $t = 0$ and $\alpha = \beta = 1$. In the animation plot you should display the streamlines at different times, as well as the full trajectory of the particle. Discuss the relation between the streamlines and the trajectory of the particle.

2.1 Streamlines and Pathlines